

Filament notification bell & Reverb — session notes

Documentation of the assistant responses for the **last three prompts** in the deposit-notification / real-time bell workstream (May 2026).

1. Asynchronous, more frequent bell refresh

User prompt: The notification bell needs to refresh asynchronously and more frequently to show new notifications.

Summary

The bell was updated so it polls more often, loads the badge count on page load, and refreshes immediately after relevant admin actions on the same page.

Polling (default 5 seconds)

- **Tenant** and **member** panels use `databaseNotificationsPolling()` driven by config (initially **5s**; Filament’s default was ~30s).
- Tunable via `FILAMENT_DATABASE_NOTIFICATIONS_POLLING` in `.env` or `config/fundflow.php` (`database_notifications_polling`).

Faster first load

- `databaseNotifications(isLazy: false)` so the **badge count** appears when the page loads, not only after opening the bell.

Instant refresh after actions

- After deposit/cash-out accept/reject on admin list pages, the bell refreshes via Livewire (`$refresh()` on the `DatabaseNotifications` component), same pattern as insights widgets.
- A listener on `NotificationSent` (database channel) can dispatch Filament’s `DatabaseNotificationsSent` when broadcasting is enabled later.

Key files

File	Role
<code>app/Filament/Support/DatabaseNotificationsRefresh.php</code>	Polli interval helper + dispatch refresh/broadcast
<code>config/fundflow.php</code>	<code>FILAMENT_DATABASE_NOTIFICATIONS_POLLING</code>
<code>app/Providers/Filament/TenantPanelProvider.php</code>	Panel bell config
<code>app/Providers/Filament/MemberPanelProvider.php</code>	Panel bell config
<code>app/Providers/AppServiceProvider.php</code>	<code>NotificationSent</code> → broadcast hook
<code>app/Filament/Tenant/Resources/FundPostings/FundPostingsResource.php</code>	Deposit/accept/reject
<code>app/Filament/Tenant/Resources/CashOutRequests/CashOutRequestsResource.php</code>	Cash-out/accept/reject

Operational note

When a **member** submits a deposit, **admins on another browser** still depended on polling (up to ~5s) until Echo/Reverb was enabled (see prompt 2).

2. Enable Filament broadcasting / Echo + 10s polling

User prompt: Enable Filament broadcasting/Echo and set `FILAMENT_DATABASE_NOTIFICATIONS_POLLING=10s` in `.env`.

Summary

Laravel Reverb was installed and wired for Filament's Echo client. Polling was set to **10 seconds** as a fallback when websockets are unavailable.

Configuration

- `composer require laravel/reverb`
- Published: `config/broadcasting.php`, `config/reverb.php`, `config/filament.php`, `routes/channels.php`
- `config/filament.php` — Echo client with `broadcaster: reverb` and `VITE_REVERB_*` env vars
- `bootstrap/app.php` — `withBroadcasting()` for `/broadcasting/auth` with Filament session auth + tenancy middleware (`InitializeTenancyByDomainEarly`, etc.)
- `routes/channels.php` — private channels:
 - `App\Models.Tenant.User.{id}`
 - `App\Models.Central.User.{id}`
- `App\Events\DatabaseNotificationsSentNow` — extends Filament's `DatabaseNotificationsSent` with `ShouldBroadcastNow` (no queue delay for bell updates)
- `.env` / `.env.example`:
 - `BROADCAST_CONNECTION=reverb`
 - `FILAMENT_DATABASE_NOTIFICATIONS_POLLING=10s`
 - Reverb app credentials, public host (`REVERB_HOST`), server bind (`REVERB_SERVER_HOST=127.0.0.1`, `REVERB_SERVER_PORT=8080`)
 - `VITE_REVERB_*` for frontend build

Production services required

1. **Reverb process** (persistent) — use Supervisor program `fundflow-reverb` (see [docs/reverb-supervisor.md](#)):
`sudo supervisorctl start fundflow-reverb`
Do not leave a manual `php artisan reverb:start` running in SSH alongside Supervisor.
2. **Reverse proxy** — Browsers connect to `wss://<host>:443`; Reverb listens on `127.0.0.1:8080`. Nginx (or similar) must proxy WebSocket traffic to that port.
3. **Queue worker** — Still needed for other queued jobs (`QUEUE_CONNECTION=database`). Bell pushes use `ShouldBroadcastNow` and do not wait on the queue.
4. **Frontend** — `npm run build` (or `dev`) so `VITE_REVERB_*` are baked into assets.

Verify

After `php artisan config:clear`, load admin/member panel and check the browser console — Echo should connect when Reverb is running. If websockets fail, the bell still refreshes every **10 seconds** via Livewire polling.

Tests

`tests/Feature/Tenant/MemberPortalTest.php` — asserts member panel has database notifications, non-lazy load, 10s polling, broadcasting enabled, Reverb broadcaster in config.

3. `reverb:start` — Class "Redis" not found

User prompt: Terminal output from `php artisan reverb:start` failing with:

```
Class "Redis" not found
at vendor/laravel/framework/.../PhpRedisConnector.php:80
```

Root cause

Laravel booted with `CACHE_STORE=redis` and `REDIS_CLIENT=phpredis`, but the **PHP redis extension** was not installed. Reverb's start command uses the cache driver early (e.g. graceful-restart via `Cache::get('laravel:reverb:restart')`), so the process failed before binding to the WebSocket port.

This is separate from:

- The **Redis server** (daemon on port 6379)
- Reverb's **react-redis** client used for horizontal scaling pub/sub when `REVERB_SCALING_ENABLED=true`

Stancl tenancy also expects Redis for **tagged tenant cache** when `CacheTenancyBootstrapper` is active (see `.env.example` comment on `CACHE_STORE=redis`).

Fix applied on server

1. Installed `php8.4-redis` and `redis-server`
2. Verified: `php -m` lists `redis`; `redis-cli ping` → PONG
3. Ran `php artisan config:clear`

After that, `reverb:start` passed the Redis bootstrap error.

Follow-up: port already in use

Next attempt reported:

```
Failed to listen on "tcp://127.0.0.1:8080": Address already in use
```

An existing process was already running `php artisan reverb:start` on port **8080**. No second instance is needed unless you intentionally restart.

Restart options:

```
php artisan reverb:restart    # signal running server
# or stop Supervisor unit / kill PID, then:
php artisan reverb:start
```

Recommendations

Setting	Recommendation
<code>REVERB_SCALING_ENABLED</code>	Leave false unless running multiple Reverb nodes
<code>CACHE_STORE=redis</code>	Keep only if <code>php8.4-redis</code> + <code>redis-server</code> are installed
Nginx	Proxy WebSockets to 127.0.0.1:8080
Assets	Rebuild with <code>npm run build</code> so Echo uses <code>VITE_REVERB_*</code>

Related work (same session, earlier prompts)

For context, these changes preceded the three prompts above:

- **Readable deposit notifications** — `FundPostingNotificationFormatter`, HTML sections, `dir="auto" / bidi` for Arabic names (`resources/css/filament/mobile-panels.css`).

- **Route fix** — Admin notification URLs use `panel: 'tenant'`; member URLs use `panel: 'member'` (avoids `filament.member.resources.fund-postings.index` not defined).
 - **Deposit notifications** — Admins on new deposit; members on accept/reject with settlement breakdown; Filament format: `filament` required for bell visibility.
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Quick reference — env vars

```
BROADCAST_CONNECTION=reverb
FILAMENT_DATABASE_NOTIFICATIONS_POLLING=10s
```

```
REVERB_APP_ID=...
REVERB_APP_KEY=...
REVERB_APP_SECRET=...
REVERB_HOST=fundflow-saas.osamman.com
REVERB_PORT=443
REVERB_SCHEME=https
REVERB_SERVER_HOST=127.0.0.1
REVERB_SERVER_PORT=8080
REVERB_SERVER_SCHEME=http
```

```
VITE_REVERB_APP_KEY="${REVERB_APP_KEY}"
VITE_REVERB_HOST="${REVERB_HOST}"
VITE_REVERB_PORT="${REVERB_PORT}"
VITE_REVERB_SCHEME="${REVERB_SCHEME}"
```

```
CACHE_STORE=redis
REDIS_CLIENT=phpredis
REDIS_HOST=127.0.0.1
REDIS_PORT=6379
```

Requires **php8.4-redis** extension and **redis-server** when using Redis for cache.